

FULL ARTICLE

Inter-regional networks and productive efficiency in Japan

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Abstract

Traditional agglomeration theories cannot explain Japan's sustained reduction of regional disparities. This study employs stochastic frontier analysis to investigate the role of inter-regional networks in improving productive efficiency using the concept of "borrowed size." Improved inter-regional networks are a driving force for strengthening local economies and converging regional disparities. Borrowed size from large city agglomeration was highly effective in local areas. The development of transportation networks has enabled local areas to catch up to large metropolitan areas. The results suggest that the development of high-speed transportation networks can help to reduce regional disparities in the Japanese economy.

KEYWORDS

agglomeration, inter-regional network, Japan, regional disparities, stochastic frontier analysis

JEL CLASSIFICATION

R10; R11

1 | INTRODUCTION

In Japan, the concentration of economic activities in Tokyo is increasing. Tokyo has economic advantages, enabling it to overcome high land prices and congestion costs. Thus, Tokyo exhibits a high level of agglomeration of economic activities. The expansion of city size through such centralization increases the productivity and productive efficiency of cities and achieves sustained growth. The role of agglomeration in improving productivity and productive efficiency is well known in the context of agglomeration economies (Marshall, 1890). Traditional empirical studies have revealed



the importance of economies of scale and transportation costs as factors that are conducive to the creation of agglomeration economies (Combes & Gobillon, 2015; Rosenthal & Strange, 2004). Geographic links between firms facilitate research and development and innovation through exchange of ideas, information, and technological knowledge (Carlino & Kerr, 2015; Fujita & Thisse, 2002). An advantage of agglomeration is that geographic proximity to the centralized location of firms increases and improves the information available to all market participants. Through the process of information interchange, market participants can develop more appropriate and accurate forecasts of the market environment and improve their ability to compete in the market. Knowledge accumulated by a firm is transmitted to other firms in various ways. This knowledge spillover leads to the development of new technologies and products, thereby increasing competitiveness in the region.

These traditional agglomeration theories predict the permanent growth of large cities at the expense of local (peripheral) areas. Furthermore, if agglomeration's lock-in effect works, the disparities between large cities and local areas will inevitably expand. In Japan, however, the concentration of economic activities in Tokyo has accelerated since the mid-1990s, but regional disparities have not increased (see Appendix A). Rather, the economic power of local areas is catching up, and economic disparities between large cities and local areas are narrowing (Otsuka & Goto, 2016).¹ Traditional agglomeration theory cannot explain these realities satisfactorily.

One potential explanation is the scope of agglomeration economies. High-speed transportation networks have been developed nationwide in Japan, with Tokyo as the hub of the network. The formation of inter-regional networks as a result of investment in transportation infrastructure has expanded the scope of innovation and knowledge spillovers in agglomeration areas. The development of high-speed railways and airports reduces the time costs of travel and increases opportunities for people to come into contact. This development not only increases the productive efficiency of a particular firm but also increases that of other firms by increasing the frequency of information exchange. Firms that are not located in agglomeration areas can increase productive efficiency by easily accessing the agglomeration economies of agglomeration areas. The knowledge spillover beyond the region suggests that the scope of the agglomeration economies expands. The economic impact of expanding agglomeration geographic coverage through transportation networks has been verified by many studies for many countries (Graham, 2007; Holl, 2012; Lall, Shalizi, & Deichmann, 2004; Melo, Graham, Levinson, & Aarabi, 2016; Montolio & Solé-Ollé, 2009; Rice, Venables, & Patacchini, 2006; Stelder, 2016). However, research on Japan, to date, has been insufficient.

This study analyses the impact of agglomeration and inter-regional networks on the productive efficiency of the regional economy. Several studies have identified the relationship between agglomeration and productive efficiency but they have tended to focus on specific economic activities rather than on the entire national economy (Beeson & Husted, 1989; Driffield & Munday, 2001; Fritsch & Slavtchev, 2011; Ke & Yu, 2014; McCoy & Moomaw, 1995; Mitra, 1999, 2000; Tsekeris & Papaioannou, 2017). Otsuka, Goto, and Sueyoshi (2010) and Otsuka and Goto (2013) show that agglomeration and improved inter-regional networks have a positive impact on the productive efficiency of the Japanese industries. This finding indicates that productive efficiency increases with decreasing inter-regional distances to large cities. However, decreasing distances to large cities also has a negative effect of increasing inter-regional competition. This is described as "agglomeration shadows" in the new economic geography (NEG) theory. This study clarifies whether positive or negative effects of the inter-regional network dominate by using Japan's latest data and a renewed market access index.

In particular, previous studies have not examined the impact of inter-regional networks on regional disparities of productive efficiency at all. In Japan, certain large metropolitan areas, such as the Greater Tokyo area and Kansai, are densely populated urban centres. On the one hand, in these large metropolitan areas, the agglomeration force is strengthened, and there are large agglomeration economies. On the other hand, in many local areas other than large cities, population density is declining, and agglomeration force is weakening. In local areas, however, improved transportation infrastructure has reduced transportation costs and attracted investment in factories. This has enabled the vertical division of the manufacturing industry across the country and helped to achieve high cost-efficiency. In addi-

¹Otsuka and Goto (2016) confirms the convergence of regional disparity by using the stochastic convergence model.



tion, local areas have succeeded in shortening the travel time to large cities through the development of high-speed transportation networks, and local areas have achieved strong economic growth by “borrowing” agglomeration economies from large cities. This study contributes to the literature by providing new findings on whether regional disparities have narrowed due to the ability of local areas to borrow the agglomeration economies of large cities.

Furthermore, this study focuses on the differences between industries in inter-regional network effects. Traditionally, for regional economies, the manufacturing industry has been regarded as the basic (export) industry, and the non-manufacturing industry has been regarded as the non-basic industry (McCann, 2001). Because of the relatively large number of transactions outside the region in the manufacturing industry, which is the export industry, product planning, development, procurement, and delivery cycles are likely to take place across regions. However, since the non-manufacturing industry is centred on the local service sector, economic activity across regions may be less frequent than for the manufacturing industry. This study clarifies whether the economic effects of inter-regional networks in the manufacturing industry outweigh those in the non-manufacturing industry.

The analysis of this study on inter-regional networks has great policy significance. Japan's national land plan aims to make cities more compact and to strengthen intercity networks as a means to counter the shrinking economy caused by population decline (see Appendix B for a detailed description of Japan's national land plan). The government aims to achieve sustainable growth by forming a “super-megaregion,” in which several large metropolitan areas integrate into a single economic area, and by dealing with contracting economies using the large agglomeration effects generated there. However, there is no empirical evidence to support this type of government policy. Therefore, this study provides valuable insights into the validity of government policies on inter-regional networking.

The structure of the rest of this paper is as follows. Section 2 provides theoretical background on inter-regional networks. Section 3 describes the methods of analysis. Section 4 presents the analysis results and interprets them. Section 5 concludes and discusses policy implications.

2 | RELATED STUDIES: THEORETICAL BACKGROUND

In traditional agglomeration theory, transportation cost and economies of scale are the primary forces shaping urban functions and spatial organizations. Thus, urban scale is characterized by high efficiency, and higher efficiency is considered to lead to higher growth. However, it is impossible for the traditional theory to explain the reality that local areas' growth exceeds that of large cities and is catching up to that of large cities. The city network theory, by contrast, solves this contradiction (Camagni, 1993). The city network theory introduces the notion that cities can borrow the functions of other cities, not only because of their physical proximity to other cities but also because of their horizontal and non-hierarchical relationships (Boix & Trullen, 2007; Camagni, 1993; Camagni & Capello, 2004; Capello, 2000). Known as “urban network externalities,” this concept has been demonstrated in the literature (Boix & Trullen, 2007; Camagni, 1993; Hall & Pain, 2006).

The starting point for this concept is agglomeration theory, which is developed on the basis of Alonso's (1973, p., 200) concept of “borrowed size” (Burger, Meijers, Hoogerbrugge, & Tresserra, 2015; Meijers, Burger, & Hoogerbrugge, 2016). The traditional concept of borrowed size mixes demographics, functionality, and critical mass effects for demand and supply and in small cities, “people can use the shopping and entertainment facilities of other cities to complement their own, businessmen can share such facilities as warehousing and business services, and labour markets enjoy a wider and more flexible range of demand and supply” (Alonso, 1973, p.200). Thus, agglomeration theory purports that small cities can borrow some of the benefits of agglomeration, while avoiding congestion. This approach underscores the fact that the agglomeration effects of cities are not necessarily limited to their physical boundaries but may spread to surrounding areas. Thus, it can explain why local areas are more efficient than larger cities and why efficient urban structures exist in a local area where agglomeration economies are borrowed from the entire urban system. The concepts of “externality fields” (Phelps, Fallon, & Williams, 2001) and “regional externalities” (Parr, 2002) were proposed to highlight the spatial scope of urban advantage that extends far beyond city boundaries.



Traditionally, agglomeration economies have been described in the context of urban economics (Fujita & Thisse, 2002). The agglomeration of population and firms explains the cumulative development of urban economies. Expansion of the scope of agglomeration means that the external economies of urban agglomeration, which have been limited to urban spaces up to now, expands widely beyond the boundaries of cities. In other words, the concept of borrowing size has the potential to be one basis for explaining the development of the entire regional economy beyond cities.² Put differently, the theoretical concept of borrowed size can overcome the limitations of traditional geographic approaches, which assume that only cities adjacent to large cities accrue efficiency gains. The concept of borrowed size also overcomes the limitations of geographic approaches that assume size is the only determinant of productive efficiency and agglomeration economies. Incorporating long-distance cooperation agreements across the regional network can improve a region's productive efficiency, even if it is a small local region.

According to Camagni, Capello, and Caragliu (2015, 2016), the traditional measure of borrowed size is calculated as the population of neighboring cities discounted by distance:

$$\text{borrowed size}_c = \sum_{j=1}^n \frac{\text{pop}_j}{w_{c,j}}, \forall c \neq j, \quad (1)$$

where c and j are different cities. w is the distance matrix representing the spatial interdependence between all cities. POP is the urban population. The borrowed size of cities is a function of the population and the distance of neighboring cities. In other words, the more neighboring populations increase, the larger the borrowed size becomes. In addition, the shorter the distance to large cities, the greater the borrowed size is.

This traditional borrowed size measure (1) emphasizes positive urbanization economies through connectivity in networks. However, it should also be noted that network connectivity can have negative economic effects because of competitive processes. This negative economic impact is captured by the notion of agglomeration shadows, which is captured by NEG theory. NEG models predict the shadow effects of surrounding agglomeration (Dobkins & Ioannides, 2001). The negative effects of competition through network connectivity are well known in Japan as the "pull-in phenomenon." In other words, there are positive and negative effects on the effects of the inter-regional network. When the positive effects outweigh the negative effects for a region, that region enjoys the economic benefits of regional network formation. This study clarifies the net impact of agglomeration areas on inter-regional networks.

3 | METHOD

3.1 | Stochastic frontier analysis

This study undertakes a stochastic frontier analysis (SFA) to estimate the effects of agglomeration and inter-regional network on productive efficiency. The SFA employed in this study conforms to the stochastic frontier function conditions proposed by Aigner, Lovell, and Schmidt (1977). Furthermore, this study utilizes the one-step approach of Battese and Coelli (1995). It simultaneously estimates the production frontier function and the productive inefficiency term's determinant. Traditionally, a two-step estimation method is used, in which inefficiency is first obtained by estimation of the stochastic frontier function, and the value of inefficiency is subsequently regressed to the determinant. However, in this case, there is a discrepancy between the hypothesis regarding the distribution of the stochastic frontier function's inefficiency term and the hypothesis regarding the regression equation that analyses the inefficiency determinant. Thus, consistency of estimation results cannot be guaranteed under the two-step approach (Coelli, 1995). By adopting the one-step approach, we are able to avoid this problem. This SFA model approximates the eco-

²Japan's latest national land plan aims to achieve sustainable regional development by utilizing urban agglomeration economies. By forming inter-regional networks, the plan is aimed at expanding agglomeration economies' power, which has been limited to large metropolitan areas, to local areas, and to achieve sustainable economic growth. See Appendix B for further details.



conomic productive efficiency level based on a non-negative error term on one side.

In this study, as a production frontier function, we formulate the translog production function as follows:

$$\ln Y_{jt} = \alpha_L \ln L_{jt} + \alpha_K \ln K_{jt} + \alpha_T T + \frac{1}{2} \alpha_{LL} (\ln L_{jt})^2 + \frac{1}{2} \alpha_{KK} (\ln K_{jt})^2 + \frac{1}{2} \alpha_{TT} T^2 + \alpha_{LK} \ln L_{jt} \ln K_{jt} + \alpha_{LT} \ln L_{jt} T + \alpha_{KT} \ln K_{jt} T + \alpha_j + v_{jt} - u_{jt}, \quad (2)$$

where j is a region, t is time, Y is real output, L is labour input, K is capital input, and T is the time trend variable representing technological progress. α is the estimation parameter. Fixed effects α_j are considered to control the characteristics of each prefecture that cannot be captured by explanatory variables.

The error term ($v_{jt} - u_{jt}$) consists of two parts, a random error term v_{jt} and an error term on productive efficiency u_{jt} . v_{jt} follows a normal distribution $N(0, \sigma^2)$ and is assumed to be independent of u_{jt} and all explanatory variables. It is assumed that u_{jt} is a non-negative stochastic variable and follows a normal distribution $N(\mu, \sigma_u^2)$.

Under the specification of (2) given, the score of productive efficiency, TEF_{jt} is defined as the ratio of the observed output Y_{jt} to the output y_{jt} on the production frontier, as follows:

$$TEF_{jt} = Y_{jt}/y_{jt} = e^{-u_{jt}}.$$

Furthermore, the following equation explains the average μ of production inefficiency:

$$\mu_{jt} = \delta_0 - \delta_{DEN} \ln DEN_{jt} - \delta_{ACC} \ln ACC_{jt}, \quad (3)$$

where δ is an estimation parameter. DEN is population density, which is defined as the population per habitable unit of land area, and represents the strength of agglomeration forces in the region. ACC is a market accessibility index that shows the strength of an inter-regional network. These variables are also used as a measure of the regional agglomeration and market potential in Otsuka et al. (2010), Fritsch and Slavtchev (2011), and Tsekeris and Papaioannou (2017).

To facilitate readers' understanding, we define the determinants of inefficiency as negative. In other words, if the determinants improve productive efficiency, δ_{DEN} and δ_{ACC} become positive³; that is, the determinants work to reduce the average of inefficiency μ . When the population density increases, this can have a positive effect on productive efficiency due to agglomeration economies. Strengthening the inter-regional networks by investing in transportation infrastructure also increases productive efficiency. However, if the "agglomeration shadow" manifests, the growth of inter-regional networks may lower production efficiency. Therefore, the impact of market access on productive efficiency yields both positive and negative effects.

As an indicator of the strength of the inter-regional network, the ACC index is calculated as follows:

$$ACC_{jt} = \sum_{k \neq j} \left(\frac{POP_{kt} / \sum_{k \neq j} POP_{kt}}{T_{jkt}} \right), \quad (4)$$

where population POP_{kt} represents the market scale of region k over period t . T_{jkt} is the distance resistance (travel time) to travel from region j to region k over period t . The travel time is obtained by calculating the weighted average of the share of time required for air, rail, and automobile travel for all combinations of departure place and destination. For each combination, the shortest amount of time to reach a destination when traveling by air, rail, or automobile is included in the calculation.⁴ Many previous studies evaluate T based on physical distance (Batty, 2009; Camagni

³In previous studies (Fritsch & Slavtchev, 2011; Otsuka et al., 2010; Tsekeris & Papaioannou, 2017), the signs for population density and the market access indicator are shown as negative, because each variable reduces inefficiency. In this study, both variables are originally defined as negative to facilitate readers' understanding. Therefore, we expect the signs of δ_{DEN} and δ_{ACC} to be positive. In other words, if the signs of δ_{DEN} and δ_{ACC} are positive, as expected, the impact of each variable on productive inefficiency will be negative, and in that sense, our setting does not contradict that of previous studies.

⁴See MLIT (2017) for further details.



et al., 2016; Hansen, 1959; Wetwitoo & Kato, 2017). On the contrary, the ACC index (4) evaluates T by travel time for the distance. In other words, the distinguishing feature of this index is that inter-regional networks are not limited to the relative market scale of the other regions, but also consider the travel time needed to access other regions. That is, the market potential of a region is evaluated by discounting the relative size of markets in other regions by the travel time for the distance between regions. This index increases as the travel time for a distance between regions decreases. If the travel time to large market regions decreases, this index increases significantly. This means that the market potential of this region is increasing.

Based on the model structure above, the estimated level of productive efficiency TEF_{jt} becomes a non-negative value with an upper limit of 1. In other words:

$$TEF_{jt} = E(\exp(-u_{jt}) | v_{jt} - u_{jt}), 0 < TEF_{jt} \leq 1.$$

Given this relationship, the change rate of productive efficiency is represented by the following equation:

$$\dot{TEF}_{jt} = \delta_{DEN} \dot{DEN}_{jt} + \delta_{ACC} \dot{ACC}_{jt}, \quad (5)$$

where the dot over each variable represents the percentage change of that variable. Changes in productive efficiency can be decomposed into: (i) changes in population density; and (ii) changes in market access.

3.2 | Convergence of productive efficiency

The presence or absence of inter-regional disparity convergence in productive efficiency can be verified by regressing the change rate of productive efficiency on the initial level of productive efficiency. We can determine this by observing the regression coefficient known as the convergence coefficient (Fukao & Yue, 2000; Otsuka, 2017). This is represented in the following equation:

$$\dot{TEF}_{jt} = \alpha + \beta \cdot TEF_{j0} + \varepsilon_j. \quad (6)$$

The left side in (6) is the annual average rate of change of productive efficiency from the initial point in time 0 until period t . When the regression coefficient β is negative, inter-regional disparity has been reduced. Conversely, if β is positive, inter-regional disparity has increased. The size of the regression coefficient β represents the speed of the convergence or divergence.

When TEF is represented as e , the regression coefficient β is represented by the definition of the following regression equation:

$$\beta = \frac{\sum_j (e_{j0} - \bar{e}_0) \cdot (\dot{e}_j - \bar{\dot{e}})}{\sum_j (e_{j0} - \bar{e}_0)^2}.$$

Since the annual average rate of change of productive efficiency is expressed by (5), we conveniently replace each term on the right side of (5) with the following equation:

$$\dot{e}_j = \dot{e}_j(DEN) + \dot{e}_j(ACC). \quad (7)$$

Based on (7), when calculating the covariance between the average rate of change of productive efficiency and the initial productive efficiency, we obtain the following equation:

$$\text{cov}(e_{j0}, \dot{e}_j) = \frac{\sum_j (e_{j0} - \bar{e}_0) (\dot{e}_j - \bar{\dot{e}})}{n} = \text{cov}(e_{j0}, \dot{e}_j(DEN)) + \text{cov}(e_{j0}, \dot{e}_j(ACC)).$$

If we divide both sides by the variation of productive efficiency at the initial time, we obtain the following formula:



$$\frac{\sum_j (e_{j0} - \bar{e}_0) (\dot{e}_j - \bar{\dot{e}})}{\sum_j (e_{j0} - \bar{e}_0)^2} = \frac{\sum_j (e_{j0} - \bar{e}_0) (\dot{e}_j(DEN) - \bar{\dot{e}}(DEN))}{\sum_j (e_{j0} - \bar{e}_0)^2} + \frac{\sum_j (e_{j0} - \bar{e}_0) (\dot{e}_j(ACC) - \bar{\dot{e}}(ACC))}{\sum_j (e_{j0} - \bar{e}_0)^2}.$$

Since the value on the left side is the regression coefficient β , replacing the variables for each term on the right side again yields the decomposition formula of the convergence coefficient, as follows:

$$\beta = \beta_{DEN} + \beta_{ACC}. \quad (8)$$

Each term on the right side is a coefficient that regresses each growth contribution on the level of its initial productive efficiency.⁵ If the convergence contribution coefficient β_i ($i = DEN, ACC$) is positive, the element's change increases the inter-regional disparities of productive efficiency. By contrast, if the convergence contribution coefficient β_i is negative, the element's change reduces the inter-regional disparities of productive efficiency. In addition, if $\beta_{DEN} + \beta_{ACC} < 0$ is realized even for $\beta_{DEN} > 0$ and $\beta_{ACC} < 0$, this can be interpreted as a reduction of inter-regional disparities in all of the determinants of productive efficiency.

4 | RESULTS

Data used in the analysis are panel data for Japan's 47 prefectures from 1980 to 2014. A description of the data source is given in Appendix C. The software used for the estimation is Frontier 4.1 (Coelli, 1996). In this software package, stochastic frontier models with fixed effects can be estimated for panel data. Table 1 presents the results of the maximum likelihood estimation using (2) and (3). The estimation equation for the determinants of efficiency is estimated at the same time as the production frontier function. In this study, two models are estimated from the perspective of introducing time effects in terms of population density and market access. In models A, C, and E, no time effects are introduced, while in models B, D, and F, time effects are introduced.

To determine whether the deviations from the estimated frontier were due to the effects of inefficiency, the null hypothesis $\gamma = 0$ was tested against the alternative hypothesis $\gamma > 0$. The null hypothesis was rejected at the 1% significance level and γ was statistically distinct from zero. This suggests that there is an efficiency effect, and thus, we confirm the estimates of the variables on the determinants of efficiency. There is no significant difference in the estimation results with or without consideration of the time effect. Models B, D, and F are selected from the log likelihood sizes, and the results are evaluated as follows.

The coefficient sign of population density and market access, which are variables explaining productive efficiency, are positive and statistically significant. This result indicates that agglomeration and inter-regional networks positively influence productive efficiency. In both the manufacturing and non-manufacturing industries, market access parameters exceed population density parameters. In particular, the market access parameters are larger in the manufacturing industry than in the non-manufacturing industry. This indicates that the borrowed size impact of inter-regional networking is greater in the manufacturing industry than in the non-manufacturing industry. In other words, in terms of passenger travel, the effects of inter-regional networks are large in the manufacturing industry. This result is as expected in this study. In other words, this result implies that the geographical scope of production activities in the manufacturing industry extends beyond the region and further, suggesting that the manufacturing industry is the basic (export) industry of the regional economy.

Table 2 shows the productive efficiency values for 11 regions in Japan. The productive efficiency values for each region are the averages of prefectures within each region. The region with the highest productive efficiency is the Greater Tokyo area, where efficiency is 0.989 (close to 1) for all industries. Productive efficiency tends to be higher in large metropolitan areas, such as Kansai (0.969) and Chubu (0.923); Hokkaido exhibits the lowest level of productive efficiency at 0.809. This is probably due to the increasing dispersion of the population and the difficulty of

⁵Fukao and Yue (2000) and Otsuka (2017) apply this method in analysing regional disparities in labour productivity.



TABLE 1 Estimation results

	All industries			Manufacturing industry			Non-manufacturing industry	
	A	B	C	D	E	F		
Production frontier function								
Constant (α_0)	-0.134 (0.063) **	-0.064 (0.031) **	-0.157 (0.067) **	0.177 (0.036) **	0.155 (0.036) **	0.501 (0.039) **		
α_L	0.656 (0.032) **	0.712 (0.053) **	0.409 (0.113) **	0.407 (0.034) **	0.368 (0.057) **	0.419 (0.049) **		
α_{LL}	0.244 (0.116) **	0.227 (0.056) **	0.329 (0.094) **	0.255 (0.038) **	0.430 (0.058) **	0.444 (0.050) **		
α_K	0.299 (0.044) **	0.195 (0.062) **	0.526 (0.140) **	0.495 (0.039) **	0.669 (0.069) **	0.602 (0.059) **		
α_{KK}	0.156 (0.132)	0.100 (0.070)	0.205 (0.147)	0.157 (0.052) **	0.522 (0.075) **	0.540 (0.065) **		
α_T	0.019 (0.000) **	0.016 (0.003) **	0.005 (0.008)	-0.010 (0.003) **	-0.011 (0.004) **	-0.041 (0.004) **		
α_{TT}	-0.001 (0.000) **	-0.001 (0.000) **	0.000 (0.000)	0.001 (0.000) **	0.001 (0.000) **	0.002 (0.000) **		
α_{LK}	-0.181 (0.150)	-0.153 (0.062) **	-0.276 (0.116) **	-0.232 (0.043) **	-0.449 (0.065) **	-0.462 (0.056) **		
α_{LT}	-0.002 (0.005)	-0.004 (0.003)	-0.001 (0.006)	-0.002 (0.002)	0.022 (0.003) **	0.021 (0.003) **		
α_{KT}	0.004 (0.001) **	0.007 (0.003) **	0.003 (0.008)	0.005 (0.002) **	-0.026 (0.004) **	-0.025 (0.003) **		
Efficiency model								
Constant (δ_0)	0.019 (0.004) **	0.075 (0.005) **	0.002 (0.004)	0.097 (0.011) **	0.101 (0.006) **	0.090 (0.005) **		
δ_{DEN}	0.028 (0.002) **	0.031 (0.001) **	0.045 (0.003) **	0.036 (0.005) **	0.033 (0.005) **	0.021 (0.001) **		
δ_{ACC}	0.045 (0.002) **	0.048 (0.002) **	0.076 (0.012) **	0.083 (0.006) **	0.053 (0.002) **	0.063 (0.002) **		
Time effect	Excluded	Included	Excluded	Included	Excluded	Included		
σ^2	0.006 (0.000) **	0.004 (0.000) **	0.020 (0.001) **	0.014 (0.000) **	0.006 (0.000) **	0.005 (0.000) **		
γ	0.090 (0.007) **	0.021 (0.003) **	0.064 (0.014) **	0.036 (0.005) **	0.016 (0.008) **	0.016 (0.003) **		
Log-likelihood	2018.39	2229.87	1017.41	1182.76	1815.21	2035.75		
Observations	1645	1645	1645	1645	1645	1645		
Mean efficiency scores	0.951	0.898	0.948	0.847	0.900	0.839		

Notes: ** and * indicate significance at the 1% and 5% level, respectively. Values in parentheses indicate standard errors.

**TABLE 2** Average efficiency score

Region	All industries	Manufacturing industry	Non-manufacturing industry
Hokkaido	0.809	0.729	0.755
Tohoku	0.842	0.770	0.782
North Kanto	0.911	0.872	0.854
Greater Tokyo area	0.989	0.971	0.938
Chubu	0.923	0.882	0.863
Hokuriku	0.871	0.806	0.808
Kansai	0.969	0.944	0.916
Chugoku	0.873	0.808	0.807
Shikoku	0.879	0.819	0.815
Kyushu	0.869	0.806	0.805
Okinawa	0.918	0.870	0.853
National average	0.898	0.847	0.839

Notes: Productive efficiency scores in the table represent average values for the period 1980–2014. The regional classification is as follows: Hokkaido (Hokkaido); Tohoku (Aomori, Iwate, Miyagi, Akita, Yamagata, Fukushima, and Niigata); North Kanto (Ibaraki, Tochigi, Gunma, and Yamanashi); Greater Tokyo area (Saitama, Chiba, Tokyo, and Kanagawa); Chubu (Nagano, Gifu, Shizuoka, Aichi, and Mie); Hokuriku (Toyama, Ishikawa, and Fukui); Kansai (Shiga, Kyoto, Osaka, Hyogo, Nara, and Wakayama); Chugoku (Tottori, Shimane, Okayama, Hiroshima, and Yamaguchi); Shikoku (Tokushima, Kagawa, Ehime, and Kochi); Kyushu (Fukuoka, Saga, Nagasaki, Kumamoto, Oita, Miyazaki, and Kagoshima); and Okinawa (Okinawa).

enjoying the benefits of inter-regional networks in regions that are not part of the Japanese mainland. The manufacturing industry has a higher level of productive efficiency than the non-manufacturing industry does. This suggests that the manufacturing industry supply chain extends to various regions in Japan and that it is easy to enjoy the benefits of shortening domestic travel time. Non-manufacturing businesses are based on inter-personal contact, which means that the non-manufacturing industry is centred on non-basic industries that support the lives of workers within the region. This suggests that people's contact across regions is less frequent than in the manufacturing industry. Furthermore, it is easy to telecommute in a non-manufacturing industry. Thus, travel time may be less relevant for some non-manufacturing businesses than for manufacturing businesses.

Table 3 shows the contributions of agglomeration and inter-regional networks in improving productive efficiency, based on Equation (5). In Hokkaido, Shikoku, and Kyushu, productive efficiency increased significantly during the observation period, exceeding the national average. The region's productive efficiency improved in both the manufacturing and non-manufacturing industries. Improving access to other markets has had a major impact on improving productive efficiency. Hokkaido's access to the mainland of Japan has improved dramatically with the increase in flights. In Shikoku, the Honshu Shikoku Bridge was completed, and access to the mainland was improved. In Kyushu, the Kyushu Shinkansen (bullet train) opened in 2004 and 2011. These effects are reflected in the results of the analysis. Tohoku, North Kanto, and Chubu have also been heavily affected by improved access to other markets. In all areas, the development of transportation infrastructure has significantly improved access to other areas. There are also some areas where the impact of population agglomeration on improving productive efficiency is evident, but this effect has been small. For example, an effect can be observed in North Kanto, the Greater Tokyo area, Chubu, Kansai, and Okinawa, but the respective effect is very small compared to the effects of market access. This indicates that the impact of inter-regional networks on improving Japanese productive efficiency is greater than the effect of the spatial agglomeration of populations.

Figure 1 plots the relationships between improving the rate of market access and improving the rate of productive efficiency for all industries. There is a clear upward-sloping relationship, which indicates a clear correlation between improved access to other markets and improved productive efficiency. Nagano and Tokushima are located in the

**TABLE 3** The growth rate of productive efficiency and factor contribution to productive efficiency

Region	All industries			Manufacturing industry			Non-manufacturing industry		
	Growth rate of efficiency score (a) + (b)	DEN (a)	ACC (b)	Growth rate of efficiency score (a) + (b)	DEN (a)	ACC (b)	Growth rate of efficiency score (a) + (b)	DEN (a)	ACC (b)
Hokkaido	0.187	-0.008	0.194	0.239	-0.010	0.249	0.182	-0.006	0.187
Tohoku	0.152	-0.012	0.163	0.194	-0.015	0.210	0.149	-0.009	0.158
North Kanto	0.148	0.005	0.142	0.190	0.007	0.183	0.142	0.004	0.138
Greater Tokyo area	0.122	0.024	0.098	0.157	0.031	0.125	0.113	0.018	0.094
Chubu	0.152	0.004	0.148	0.195	0.005	0.189	0.146	0.003	0.143
Hokuriku	0.073	-0.001	0.074	0.094	-0.001	0.096	0.071	-0.001	0.072
Kansai	0.113	0.005	0.108	0.145	0.007	0.138	0.108	0.004	0.104
Chugoku	0.107	-0.006	0.113	0.137	-0.008	0.145	0.105	-0.005	0.109
Shikoku	0.192	-0.006	0.198	0.246	-0.008	0.254	0.186	-0.005	0.191
Kyushu	0.169	-0.001	0.170	0.217	-0.001	0.218	0.164	-0.001	0.164
Okinawa	0.098	0.011	0.087	0.126	0.014	0.112	0.093	0.008	0.084
National average	0.138	0.001	0.136	0.176	0.002	0.174	0.132	0.001	0.131

Notes: Growth contribution is the value calculated by equation (5). The regional classification is as follows: Hokkaido (Hokkaido); Tohoku (Aomori, Iwate, Miyagi, Akita, Yamagata, Fukushima, and Niigata); North Kanto (Ibaraki, Tochigi, Gunma, and Yamanashi); Greater Tokyo area (Saitama, Chiba, Tokyo, and Kanagawa); Chubu (Nagano, Gifu, Shizuoka, Aichi, and Mie); Hokuriku (Toyama, Ishikawa, and Fukui); Kansai (Shiga, Kyoto, Osaka, Hyogo, Nara, and Wakayama); Chugoku (Tottori, Shimane, Okayama, Hiroshima, and Yamaguchi); Shikoku (Tokushima, Kagawa, Ehime, and Kochi); Kyushu (Fukuoka, Saga, Nagasaki, Kumamoto, Oita, Miyazaki, and Kagoshima); and Okinawa (Okinawa).

upper right. The Nagano Shinkansen has improved access to markets and increased productive efficiency in Nagano. Meanwhile, in Tokushima, the opening of the Akashi Kaikyo Bridge has improved access to the mainland, indicating that productive efficiency has increased. In this way, it can be concluded that the development of high-speed transportation networks played a role in strengthening regional economic networks and greatly increasing productive efficiency.

Finally, the presence or absence of convergence of productive efficiency is examined. Figure 2 plots the initial levels of productive efficiency and the rate of change and shows a downward-sloping tendency. This finding indicates that regions with lower productive efficiency in the initial stages had higher growth of productive efficiency than regions with higher initial productive efficiency did. This result suggests that areas with low productive efficiency may have caught up to areas with high productive efficiency. Productive efficiency in Nagano, Tokushima, and Hokkaido improved significantly, while that in Saitama, Osaka, Kanagawa, and Tokyo showed little improvement.

Table 4 shows the convergence coefficient based on (6) and (8). The convergence coefficient β is negative for both the manufacturing and non-manufacturing industries. This indicates a reduction in the regional disparities of productive efficiency during the observation period. However, the sign of the population density coefficient is positive in both industries. This means that the population agglomeration worked to expand the regional disparities of productive efficiency. By contrast, the sign of the coefficient of market access is negative, and the magnitude of the coefficient greatly exceeds the value of population density. While population agglomeration is working to increase the regional disparities of productive efficiency, market access is working to reduce the regional disparities of productive efficiency. As the latter effect exceeds the former effect, it can be concluded that the regional disparities of productive efficiency, as a whole, have diminished. These results suggest that the investment in high-speed transportation infrastructure by the Japanese

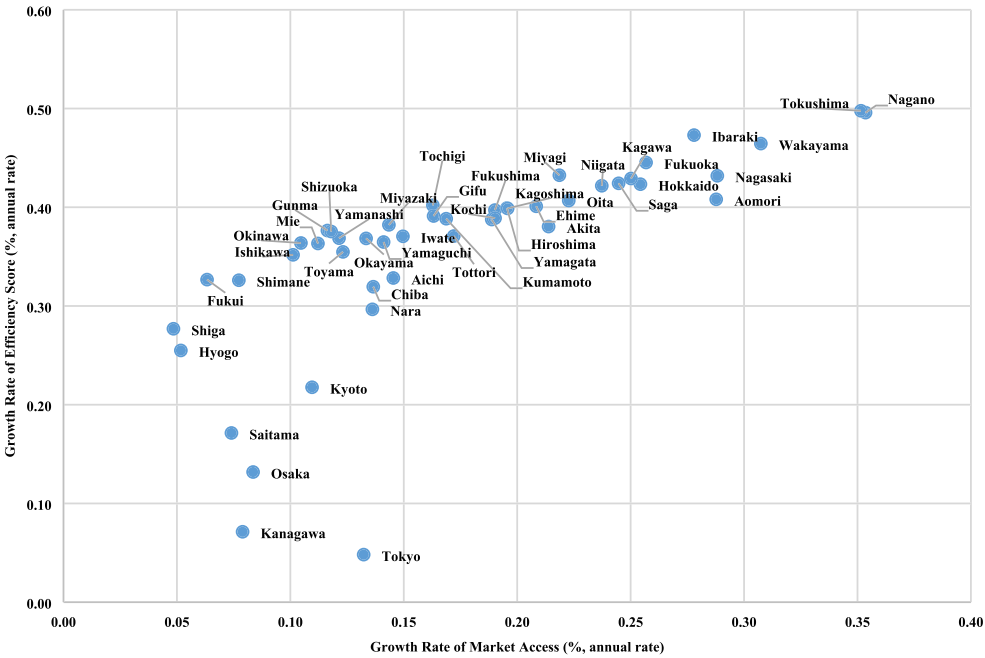


FIGURE 1 Dynamic relationship between efficiency score and market access
Note: Each point in the figure represents the value in all industries.

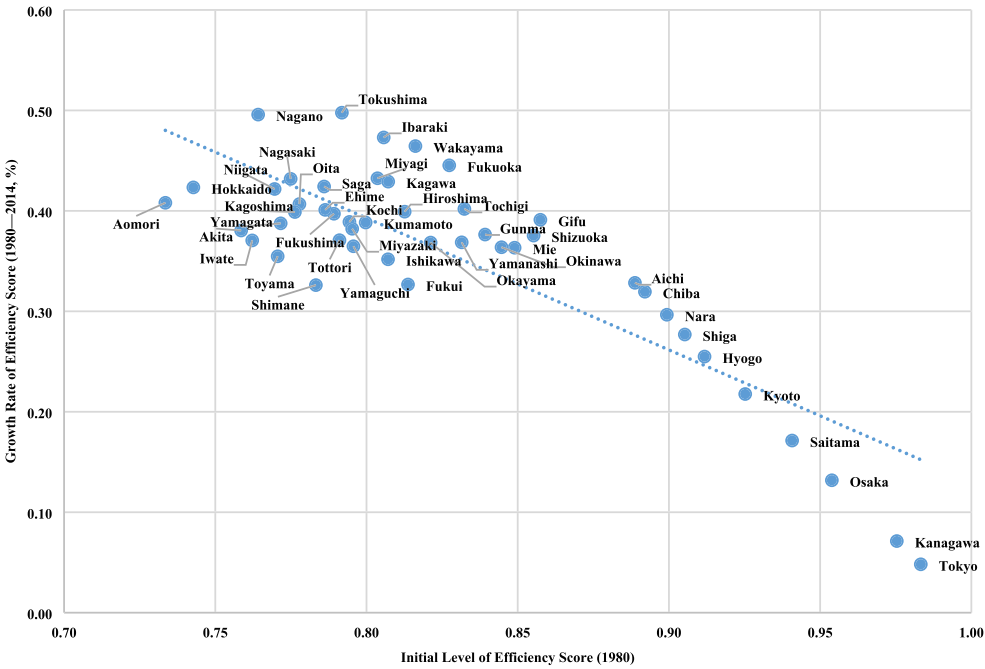


FIGURE 2 Convergence of productive efficiency
Note: Each point in the figure represents the value in all industries.



TABLE 4 Decomposition results of the beta coefficient (1980–2014)

	All industries			Manufacturing industry			Non-manufacturing industry		
Convergence coefficient (β)	-0.0029 (0.0010)	**		-0.0025 (0.0009)	**		-0.0030 (0.0009)	**	
Population density (β_{DEN})	0.0012 (0.0002)	**		0.0011 (0.0001)	**		0.0009 (0.0001)	**	
Market access (β_{ACC})	-0.0042 (0.0010)	**		-0.0036 (0.0008)	**		-0.0039 (0.0009)	**	

Notes: **, and * indicate significance at the 1%, and 5% levels, respectively.

Values in parentheses indicate standard errors.

government has strengthened local areas' economic power and helped reduce economic disparities between large metropolitan areas and local areas.

5 | DISCUSSION AND CONCLUSION

The current dynamics in Japanese regional economic systems are consistent with traditional literature, which emphasizes the importance of agglomeration economies for economic growth. Population agglomeration is progressing in the three large metropolitan areas of the Greater Tokyo area, Kansai, and Chubu, achieving high economic growth. Urbanization is relatively stable. The Greater Tokyo area's share of the population is increasing at the expense of local areas (see Appendix A). These trends follow global trends. However, there is no proof in recent studies that regional disparities have been expanding steadily. The manufacturing industry is predominantly located outside of urban areas and continues to grow, despite economic globalization, and this supports Japan's local economies. Therefore, an explanation that goes beyond the conventional agglomeration theory is required for Japan. In this study, we show that inter-regional networks enhance the effectiveness of borrowed size and complement lost links between real regional economic systems and agglomeration theory.

Network economies are paradigms that have entered a stage in which they are generally more complementary to the concepts of agglomeration economies. This study explores the roles of inter-regional networks and draws new analogies by focusing on the new roles of agglomeration economies. Agglomeration economies are known to have traditionally relied heavily on the sheer size of a region, but this study examines whether inter-regional networks complement the size of the region as a factor of regional economic growth. In particular, this study focused on upgrading passenger networks through the three major networks of roads, railways, and aviation. The empirical analysis of this study has produced new knowledge. The main findings can be summarized as follows.

First, the improvement of inter-regional networks increases the “borrowing” effect, whereby the benefits of agglomeration in large cities spills over to increase the productive efficiency of the local economy. The impact of the “borrowing size” effect is greater than the agglomeration benefit of local size. Positive effects in inter-regional networks are strongly evident, and in that sense, the effects of agglomeration shadows are not apparent.

Second, the impact of the increase in productive efficiency due to the improvement of inter-regional networks is greater for the manufacturing industry than for the non-manufacturing industry. The manufacturing industry plays a role in the basic (export) industry in regional economies. Therefore, a network with the outside area is important for production activities in the manufacturing industry.

Third, in large metropolitan areas, such as the Greater Tokyo area and Kansai, the economic benefits by region size were large. This means that the large metropolitan areas enjoyed a high level of agglomeration economies, driven by population agglomeration. If a positive lock-in effect operates in agglomeration, the large metropolitan area's sustainable growth is guaranteed.

Fourth, in local areas other than large metropolitan areas, borrowed size from large city agglomeration was highly effective in enhancing productive efficiency. The shortening of travel times to large metropolitan areas engendered by high-speed transportation networks through investment in transportation infrastructure greatly contributed to the growth of the local economy.



Fifth, the strengthening of transportation networks contributed to the convergence of regional disparities on productive efficiency. The development of transportation networks has enabled local areas to catch up to the large metropolitan areas due to the effects of borrowed size in local areas.

These findings have substantial theoretical and policy implications. Theoretically, the observation that the borrowed size effect occurs in regional networks calls for the geographic base of agglomeration theory to be re-built. It may be necessary to re-consider the appropriateness of the term “agglomeration economies” if the economic phenomenon can be shared not only within the agglomeration itself but also across regional networks. Ignoring the role of inter-regional networks means that regional development policies may be directed toward narrowly defined spatial scales. The existence of an inter-regional network economy affirms wider regional development and suggests that it can be an effective means of putting under-developed areas on a growth track.

Japan is aiming to build a giant economic zone, called a “super-megaregion,” through the formation of inter-regional networks, as defined in the national land plan. The goal is to create a national land structure that can deal with the effects of population decline. The specific aim is to achieve sustainable economic growth by shortening the travel time between regions through investment in high-speed transportation networks, namely, the Linear Chuo Shinkansen (magnetic levitation train), and increasing the borrowed size effect. The results of this study provide insights into the foundations of Japanese land policies and enable predictions of regional disparities. Specifically, the results of this study suggest that the development of high-speed transportation networks may further reduce regional disparities. This effect is particularly significant for the manufacturing industry.

Future research should deepen our consideration of these findings from a micro perspective. In other words, the mechanism of the effects of borrowed size should be elucidated. With regard to passenger travel, the number of business trips is expected to increase as inter-regional travel time decreases. However, the impact of an increase in the number of business trips on productive efficiency has not been well explored. In particular, the mechanism by which a business trip leads to increased productive efficiency needs to be clarified. Furthermore, as globalization progresses, the regional economy needs to make full use of inter-country network connectivity, and further research is needed to explore ways to utilize it. Future research should also focus on analysing and understanding networks between international regions.

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APPENDIX

REGIONAL DISPARITIES AND MIGRATION IN JAPAN

Figure A1 shows the time-series transition of the coefficient of variation (CV). In gross value added (GVA). Per unit of labour from the 1980s to 2000s. The CV rose until 1990 after having fallen sharply from 1980 to 1984. This change is attributed to the impact of the asset-inflated bubble economy in 1990. CV fell again from 1991 to 2001. During this time, the Japanese economy fell into a serious economic recession due to the collapse of the bubble economy. These 10 years are called the “lost decade.” thereafter, CV increased until 2005, but decreased again in 2008 owing to the impact of the economic downturn precipitated by the Lehman brothers bankruptcy. Thereafter, the CV rose again. While CV changed in the observation period due to the influence of economic fluctuation, the level of CV in 2014 was much lower than the level in 1980, showing a downward trend. This result implies that regional disparities of GVA per unit of labour are decreasing in the long run.

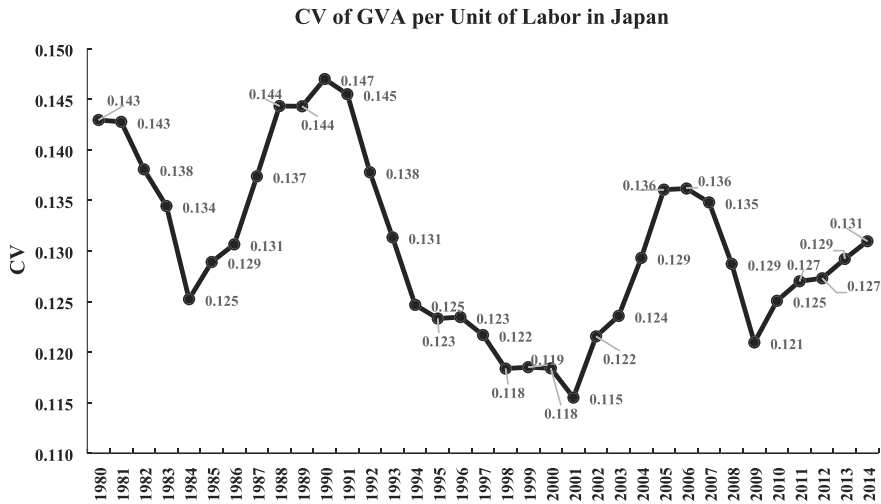


FIGURE A1 CV of GVA per unit of labour in Japan
Source: Annual report on prefectural accounts (Cabinet Office), Basic Resident Register Population (Ministry of Internal Affairs and Communications).

Figure A2 shows the net inflow of population in the greater Tokyo area, Chubu, Kansai, and local areas. A phenomenon observed in the 1980s and 2000s was the polar concentration of the population in the greater Tokyo area. In the middle of the 1990s, a reversal of population movement from this area to local areas occurred quite suddenly. However, in the 2000s, population migration from local areas to the greater Tokyo area began accelerating again. Overall, in Japan, there is a concentration of population in the greater Tokyo area.

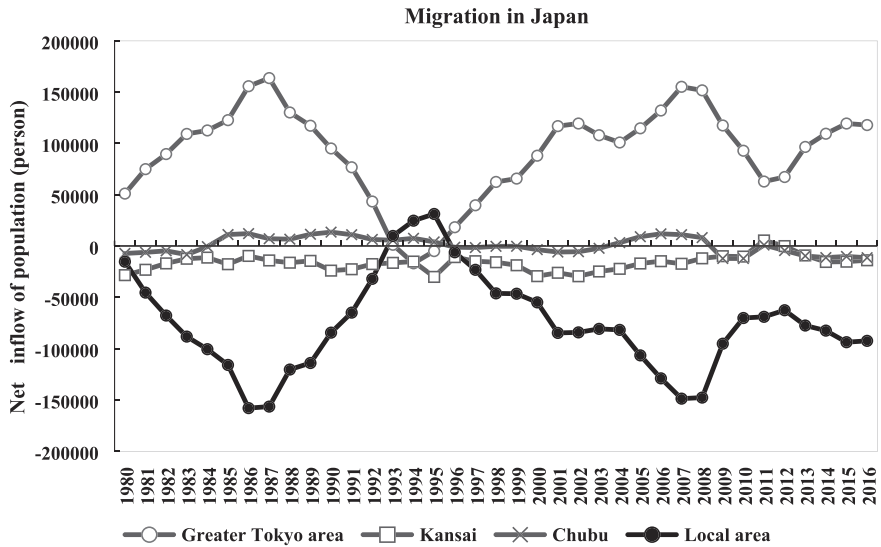


FIGURE A2 Migration in Japan
Source: Report on Internal Migration in Japan (Ministry of Internal Affairs and Communications).



APPENDIX

THE NATIONAL LAND PLAN IN JAPAN

The national land plan is a long-term, comprehensive spatial plan for the use, development, and conservation of national land. Since the formulation of the First Comprehensive National Development Plan in 1962, Japan's post-war national land planning has been based on the Comprehensive National Land Development Act. Post-war national land plans have been revised to resolve various issues in accordance with the needs of each era.

The latest national land planning period is 10 years, with the 2020 Tokyo Olympic and Paralympic Games as the interim year. Efforts during this period have been positioned to influence Japan's future development potential. The latest national land plan defines the basic concept of the plan as the creation of "a country that promotes interaction." Interaction is the inter-regional movement of people, goods, money, and information arising from the mutual cooperation of various regions with diverse personalities. It is believed that dynamic interaction between regions enables sustainable development. The means of creating this interaction is the "compactization" of the region and the formation of "inter-regional networks."

In order to provide services efficiently although the population decline defines the economy, regional integration (downsizing) is essential. The aim is to maintain the size of the markets by networking each region to prevent the markets from shrinking. The important issue here is the individuality of each region as a compact entity. Differences in individuality between regions arise when regions independently identify their own local resources and refine their attractiveness. This difference in individuality creates interaction between regions, and this interaction is thought to be the source of regional vitality. Through these efforts, Japan aims to create a national land structure that will increase the productivity and productive efficiency of the nation as a whole by providing high-quality services efficiently and creating new value, even in the face of population decline.

An important role in creating inter-regional interaction is the formation of inter-regional networks through high-speed transportation systems. The new national land plan aims to form a "super-megaregion" through the Linear Chuo Shinkansen, a high-speed transportation system, to gain high competitiveness by creating significant regional agglomeration economy effects. The Linear Chuo Shinkansen is a 438-kilometre high-speed transportation line connecting Tokyo, Nagoya, and Osaka. The development plan for the Linear Chuo Shinkansen was officially decided in 2011. The Linear Chuo Shinkansen is scheduled to open in 2027 between Tokyo and Nagoya, and in 2045 between Nagoya and Osaka. The superconducting magnetic levitation system (maximum speed: 505 kilometres/hour) has been adopted as a driving method. It is planned to connect Tokyo and Nagoya in about 40 minutes, and Tokyo and Osaka in about 1 hour.

The opening of the Linear Chuo Shinkansen will bring about a significant reduction in travel time. With this shortened travel time, Japan's major large metropolitan areas will be able to work closely together as an integrated economic zone, creating a world-class "super-megaregion" on an unparalleled economic scale. This dramatic reduction in travel time is expected to stimulate face-to-face communication between firms further. In other words, when the Linear Chuo Shinkansen connects Tokyo, Nagoya, and Osaka in approximately 1 hour, information exchanges are expected to increase, and new joint research and development and innovation may be created among firms.

APPENDIX

EXPLANATION OF DATA

The real output (Y) data of each prefecture are from the *Annual report on prefectural accounts*⁶ (Cabinet Office), and are deflated by the GDP deflator. Labour input (L) is calculated by multiplying the number of employees by the number of hours worked. The data on the number of employees in each prefecture are from the *Annual report on prefectural accounts* (Cabinet Office). Working hours data are from the *Monthly labour survey* (Ministry of Health, Labour and Welfare). The capital input (K) is the private capital stock multiplied by the capacity utilization ratio. Private capital stock data are estimated by the Central Research Institute of Electric Power Industry. The manufacturing industry capacity utilization ratio is from an index of capacity utilization ratios published by the Ministry of Economy, Trade



TABLE C1 Descriptive statistics

	Value-added (million yen)				Labour input (person-hour)				Capital input (million yen)				Market access
	All industries	Manufacturing	Non-manufacturing	All industries	Manufacturing	Non-manufacturing	All industries	Manufacturing	Non-manufacturing	Population density			
1980	Mean	7,730,508	1,372,122	6,284,633	1,364,909	295,799	1,067,572	8,496,714	3,335,672	5,248,584	1,276	22.25	
	S.D.	10,233,479	1,821,621	8,442,672	1,399,606	335,341	1,084,151	9,115,215	3,624,570	6,210,252	1,506	6.00	
	Max	62,919,057	9,063,891	53,336,575	8,410,740	1,679,754	6,729,076	45,566,369	14,405,245	36,582,428	8,372	36.14	
	Min	1,659,582	94,180	1,458,760	373,549	31,930	311,600	1,155,407	252,337	762,589	262	13.34	
1990	Mean	10,837,508	2,037,251	8,675,979	1,470,828	324,310	1,138,736	15,529,807	5,886,160	9,907,896	1,336	22.35	
	S.D.	15,406,048	2,390,362	13,204,132	1,614,268	346,780	1,281,201	17,910,196	6,215,941	13,483,438	1,574	6.16	
	Max	96,483,934	9,839,002	85,211,889	9,669,281	1,670,266	7,929,467	99,568,959	28,157,387	84,514,309	8,456	35.91	
	Min	2,123,197	148,651	1,773,178	362,099	32,312	289,755	2,681,166	482,733	1,951,542	259	12.93	
2000	Mean	10,765,104	2,011,027	8,770,252	1,372,705	254,102	1,102,140	20,111,686	7,117,994	13,695,435	1,350	25.25	
	S.D.	14,822,163	2,032,771	13,348,402	1,478,683	254,968	1,217,629	23,431,277	7,104,309	18,975,392	1,585	6.01	
	Max	93,658,397	9,050,237	86,801,664	8,715,110	1,155,040	7,370,153	131,564,767	34,653,218	117,445,951	8,413	38.51	
	Min	2,127,374	184,599	1,759,982	328,361	27,728	270,074	3,606,780	658,160	2,705,984	260	17.18	
2010	Mean	11,158,051	2,280,320	8,839,387	1,233,226	190,603	970,214	27,177,667	7,422,947	18,874,986	1,360	26.01	
	S.D.	15,275,105	2,204,401	13,654,808	1,355,706	183,762	1,067,624	32,341,314	7,345,204	27,271,084	1,679	6.11	
	Max	96,217,403	10,735,264	88,808,855	7,949,085	853,089	6,373,482	182,997,919	39,024,137	169,716,749	9,066	39.65	
	Min	1,857,687	191,663	1,591,060	279,522	25,422	227,706	4,724,266	692,392	3,654,294	249	15.86	
2014	Mean	11,662,718	2,436,275	9,150,756	1,207,253	190,814	952,281	31,586,469	7,828,993	22,145,309	1,353	26.60	
	S.D.	16,115,681	2,603,710	14,321,806	1,322,034	182,418	1,053,709	38,003,200	7,665,752	32,478,660	1,694	6.00	
	Max	101,186,156	14,521,490	93,367,789	7,733,538	890,263	6,308,599	216,663,687	41,205,418	202,599,244	9,185	39.65	
	Min	1,867,750	198,327	1,630,076	271,509	26,047	226,042	5,387,930	736,874	4,229,472	245	17.56	
1980-2014	Mean	10,341,909	1,964,520	8,355,152	1,364,108	262,363	1,078,642	18,940,610	6,133,024	12,795,671	1,343	24.16	
	S.D.	14,229,589	2,174,343	12,550,135	1,458,832	280,623	1,172,863	24,227,938	6,464,148	19,722,313	1,587	6.24	
	Max	101,828,797	14,572,817	93,616,142	9,669,281	1,707,001	7,929,467	216,663,687	43,011,704	202,599,244	9,185	40.03	
	Min	1,659,582	94,180	1,458,760	271,509	23,323	226,042	1,155,407	252,337	762,589	245	12.90	



and Industry. The capacity utilization ratio of the non-manufacturing industry is estimated using the following procedure because no official statistics are available. It is assumed that fluctuations in $\ln\left(\frac{Y}{K}\right)$ can be attributed to a change in capital utilization, as well as a time trend. Based on this assumption, a proxy of the capital utilization rate can be measured with a residual error term (ϵ). in the regression $\ln(Y/K) = \alpha + \beta T + \epsilon$, where T is a time trend and β is a time-invariant slope of $\ln\left(\frac{Y}{K}\right)$. Population density is the ratio of population to the residential area. The population data are from the *Basic resident register population* (Ministry of Internal Affairs and Communications⁷). The residential area data are from *A system of social and demographic statistics* (Ministry of Internal Affairs and Communications). In calculating market access indicators, the data for the time-distance matrix between regions are from the *National integrated transport analysis system* (Ministry of Land, Infrastructure, Transport and Tourism⁸).

The descriptive statistics are presented in Table C1. Real output increased over the course of the observation period in the manufacturing and non-manufacturing industries. After rising from the 1980s to the 1990s, the pace of increase in the 2000s was moderate. Labour input steadily decreased from 1990, while capital input increased throughout the observation period. Population density rose moderately between 1980 and 2010, and then declined slightly in 2014 owing to a nationwide decline in population. Market access indicators increased sharply from 1990, suggesting that inter-regional networks were strengthened by the opening of the bullet trains: Yamagata Shinkansen, Akita Shinkansen, Nagano Shinkansen, and Kyushu Shinkansen.

Table C2 shows the average values of population density and market access index over time by Japanese region. Population density is high in Japan's three major metropolitan areas, the Greater Tokyo area, Chubu, and Kansai. However, Hokkaido has the lowest population density. The market access index is also higher than average in the Greater Tokyo area, Chubu, and Kansai. This means that Japan's transportation network is constituted mainly of large cities. In particular, Japan has a high-speed transportation network with Tokyo as its hub. Therefore, market access indicators tend to be higher in regions with shorter travel times from Tokyo. The Okinawa market access index is higher because of the larger number of flights to Tokyo than to local areas.

TABLE C2 Regional average of population density and market access

	Population density	Market access
Hokkaido	257	17.88
Tohoku	493	19.25
North Kanto	798	25.64
Greater Tokyo area	4,619	35.93
Chubu	1,246	25.82
Hokuriku	738	20.64
Kansai	2,407	32.13
Chugoku	864	20.33
Shikoku	864	21.08
Kyushu	846	20.34
Okinawa	1,158	24.46
National average	1,343	24.16

Notes: The values in the table are the mean values of the observation period (1980–2014) for each Japanese region. The regional classification is as follows: Hokkaido (Hokkaido); Tohoku (Aomori, Iwate, Miyagi, Akita, Yamagata, Fukushima, and Niigata); North Kanto (Ibaraki, Tochigi, Gunma, and Yamanashi); Greater Tokyo area (Saitama, Chiba, Tokyo, and Kanagawa); Chubu (Nagano, Gifu, Shizuoka, Aichi, and Mie); Hokuriku (Toyama, Ishikawa, and Fukui); Kansai (Shiga, Kyoto, Osaka, Hyogo, Nara, and Wakayama); Chugoku (Tottori, Shimane, Okayama, Hiroshima, and Yamaguchi); Shikoku (Tokushima, Kagawa, Ehime, and Kochi); Kyushu (Fukuoka, Saga, Nagasaki, Kumamoto, Oita, Miyazaki, and Kagoshima); and Okinawa (Okinawa).



Resumen. Las teorías tradicionales sobre la aglomeración no pueden explicar la reducción continua de las disparidades regionales en el Japón. Este estudio emplea un análisis de frontera estocástica para investigar el papel de las redes interregionales en la mejora de la eficiencia productiva por medio del concepto de “tamaño prestado”. La mejora de las redes interregionales es una fuerza motriz para el fortalecimiento de las economías locales y la convergencia de las disparidades regionales. El tamaño prestado de la aglomeración de las grandes ciudades fue muy efectivo en las áreas locales. El desarrollo de las redes de transporte ha permitido a las zonas locales ponerse a la par de las grandes áreas metropolitanas. Los resultados sugieren que el desarrollo de las redes de transporte de alta velocidad puede ayudar a reducir las disparidades regionales en la economía japonesa.

抄録: 伝統的な集積の理論では、日本の持続的な格差の縮小を説明するのは不可能である。本稿では、確率フロンティア分析により、生産効率の改善における地域間ネットワークの役割を「借用したサイズ」の概念を用いて検討した。地域間ネットワークの改善は、地域経済を強化し、地域格差を収束させる原動力である。大都市の集積から借用したサイズは地方では高い効果を示した。交通ネットワークの開発は、地方の地域が大都市地域に追いつくことを可能にしている。結果から、高速交通ネットワークの開発が日本経済における地域間格差の縮小の一助となることが示唆される。